Our analysis also makes it clear why we assumed that there are only two variables in each clause. For if there were k, $k > 2$, variables in a clause then as any clause that is not presently satisfied may have only one incorrect setting, a randomly chosen variable whose value is changed might only increase the number of values in agreement with $\mathscr{A}$ with probability $1/k$ and so we could only conclude from our prior Markov chain results that the mean time to obtain the values in $\mathscr{A}$ is an exponential function of n, which is not an efficient algorithm when n is large. ∎

4.6. Mean Time Spent in Transient States

Consider now a finite state Markov chain and suppose that the states are numbered so that $T = \{1, 2, \ldots, t\}$ denotes the set of transient states. Let

$$\mathbf{P}_T = \begin{bmatrix} P_{11} & P_{12} & \cdots & P_{1t} \\ \cdots & \cdots & \cdots & \cdots \\ P_{t1} & P_{t2} & \cdots & P_{tt} \end{bmatrix}$$

and note that since $\mathbf{P}_T$ specifies only the transition probabilities from transient states into transient states, some of its row sums are less than 1 (otherwise, T would be a closed class of states).

For transient states i and j, let s_{ij} denote the expected number of time periods that the Markov chain is in state j, given that it starts in state i. Let $\delta_{i,j} = 1$ when $i = j$ and let it be 0 otherwise. Condition on the initial transition to obtain

$$\begin{aligned} s_{ij} &= \delta_{i,j} + \sum_k P_{ik} s_{kj} \\ &= \delta_{i,j} + \sum_{k=1}^{t} P_{ik} s_{kj} \end{aligned} \tag{4.18}$$

where the final equality follows since it is impossible to go from a recurrent to a transient state, implying that $s_{kj} = 0$ when k is a recurrent state.

Let $\mathbf{S}$ denote the matrix of values s_{ij}, $i, j = 1, \ldots, t$. That is,

$$\mathbf{S} = \begin{bmatrix} s_{11} & s_{12} & \cdots & s_{1t} \\ \cdots & \cdots & \cdots & \cdots \\ s_{t1} & s_{t2} & \cdots & s_{tt} \end{bmatrix}$$

In matrix notation, Equation (4.18) can be written as

$$\mathbf{S} = \mathbf{I} + \mathbf{P}_T \mathbf{S}$$

where $\mathbf{I}$ is the identity matrix of size t. Because the preceding equation is equivalent to

$$(\mathbf{I} - \mathbf{P}_T)\mathbf{S} = \mathbf{I}$$

we obtain, upon multiplying both sides by $(\mathbf{I} - \mathbf{P}_T)^{-1}$,

$$\mathbf{S} = (\mathbf{I} - \mathbf{P}_T)^{-1}$$

That is, the quantities s_{ij}, $i \in T$, $j \in T$, can be obtained by inverting the matrix $\mathbf{I} - \mathbf{P}_T$. (The existence of the inverse is easily established.)

Example 4.26 Consider the gambler's ruin problem with $p = 0.4$ and $N = 7$. Starting with 3 units, determine

(a) the expected amount of time the gambler has 5 units,
(b) the expected amount of time the gambler has 2 units.

Solution: The matrix $\mathbf{P}_T$, which specifies P_{ij}, $i, j \in \{1, 2, 3, 4, 5, 6\}$, is as follows:

$$\mathbf{P}_T = \begin{array}{c|cccccc} & 1 & 2 & 3 & 4 & 5 & 6 \\ \hline 1 & 0 & 0.4 & 0 & 0 & 0 & 0 \\ 2 & 0.6 & 0 & 0.4 & 0 & 0 & 0 \\ 3 & 0 & 0.6 & 0 & 0.4 & 0 & 0 \\ 4 & 0 & 0 & 0.6 & 0 & 0.4 & 0 \\ 5 & 0 & 0 & 0 & 0.6 & 0 & 0.4 \\ 6 & 0 & 0 & 0 & 0 & 0.6 & 0 \end{array}$$

Inverting $\mathbf{I} - \mathbf{P}_T$ gives

$$\mathbf{S} = (\mathbf{I} - \mathbf{P}_T)^{-1} = \begin{bmatrix} 1.6149 & 1.0248 & 0.6314 & 0.3691 & 0.1943 & 0.0777 \\ 1.5372 & 2.5619 & 1.5784 & 0.9228 & 0.4857 & 0.1943 \\ 1.4206 & 2.3677 & 2.9990 & 1.7533 & 0.9228 & 0.3691 \\ 1.2458 & 2.0763 & 2.6299 & 2.9990 & 1.5784 & 0.6314 \\ 0.9835 & 1.6391 & 2.0763 & 2.3677 & 2.5619 & 1.0248 \\ 0.5901 & 0.9835 & 1.2458 & 1.4206 & 1.5372 & 1.6149 \end{bmatrix}$$

Hence,

$$s_{3,5} = 0.9228, \qquad s_{3,2} = 2.3677$$ ∎